

ATHENA

Clock Module

Build, Verification and Learning Record

From 555/556 timing experiments to a selectable, gated CPU clock

Current status: core clock architecture functionally verified

Repository baseline and bench evidence consolidated to 18 September 2026

Scope note

Later Z80-specific frequency and clock-level adaptations are recorded separately from the original module design.

1. Purpose and scope

This document records the design, construction, testing, fault-finding and learning associated with the Athena clock module. The clock began as a practical exploration of 555-family timing behaviour and developed into a complete CPU clock subsystem with variable-speed RUN operation, manual single-step operation, persistent RUN/MANUAL mode selection, HLT gating and an observable final CPU clock.

The record deliberately separates the original clock-module architecture from later changes made only to interface that module to the 1986 NMOS Z80. It also separates confirmed bench evidence from repository baselines, deductions and items still marked for verification.

2. Executive summary

- The original learning work established astable, monostable and latch behaviour using 555-family circuits before the design was consolidated around one NE556 dual timer.
- U1A (NE556 Timer 1) generates the variable-speed astable RUN clock.
- U1B (NE556 Timer 2) generates the manual STEP pulse.
- Two gates of a SN7400 form a cross-coupled RUN/MANUAL latch; its Q output drives the selector.
- One channel of a SN74LS157 selects either the STEP pulse or the astable clock.
- The remaining SN7400 gates implement HLT gating and produce the final CPU_CLK on SN7400 pin 11.
- The NE556 was systematically verified by oscilloscope, including its threshold ramps, outputs, trigger behaviour and monostable timing node.
- A major monostable fault was traced to a $\sim 1\text{ k}\Omega$ resistor fitted where $1\text{ M}\Omega$ was required. Correcting the resistor restored the expected exponential timing ramp.
- A later end-to-end fault was traced to SN7400 pin 9 still being wired directly to NE556 pin 5, bypassing the SN74LS157 selector. Removing that legacy connection and routing SN74LS157 pin 4 to SN7400 pin 9 restored correct RUN/MANUAL operation.
- RUN, MANUAL, STEP and HLT behaviour were then functionally verified at the final CPU clock output.
- Because MANUAL/STEP produces exactly one CPU_CLK pulse per button press, a connected Z80 can later be advanced by one T-state at a time. This turns the clock into a deliberate CPU timing and bus-observation tool, not merely a slow clock source.
- For Z80 bring-up, the astable timing capacitor was later changed to 470 pF and a separate $330\text{ }\Omega$ pull-up was added at the Z80 clock input. Those are interface/operating-range adaptations, not silent changes to the original schematic baseline.

3. Design evolution

The clock work grew directly from the Ben Eater eight-bit computer learning sequence. The earlier 555-based experiments were not discarded; they established the physical meaning of the $1/3\text{ VCC}$ and $2/3\text{ VCC}$ thresholds, the discharge transistor, astable charging/discharging, monostable triggering and the distinction between an oscillator, a one-shot and a stored logic state.

The current Athena design then reorganised those functions into a more explicit architecture: generation, state, selection, gating and output. The result is not a copy of the Ben Eater clock. It retains the same educational objectives while using a dual 556 timer, TTL NAND logic and a dedicated multiplexer.

Stage	Implementation	Learning/architectural role
Foundation	Individual 555-family experiments	Understand timing thresholds, discharge switching, astable and

		monostable behaviour.
Consolidation	NE556 dual timer	Place astable and manual pulse generation in one IC while preserving observability.
Mode memory	SN7400 cross-coupled NAND latch	Store RUN versus MANUAL as a persistent logic state.
Selection	SN74LS157	Choose one of the two clock sources cleanly from the stored state.
Gating/output	SN7400 gates 3 and 4	Permit or block the selected clock under HLT control and provide CPU_CLK.

4. Architecture and signal flow

NE556 clock sources → SN7400 mode latch → SN74LS157 selector → SN7400 HLT gating → CPU_CLK

This decomposition is central to the learning value of the module. Each stage can be probed independently, and a fault can be localised to generation, stored mode, selection or final gating rather than treating the clock as a black box.

4.1 Functional roles of the three ICs

IC	Device	Function in Athena
U1	NE556 / RS 305-838	Timer 1 produces the variable astable clock; Timer 2 produces the manual STEP pulse.
U2	TI SN7400	Gates 1/2 form the RUN/MANUAL latch; gates 3/4 implement HLT gating and final clock output.
U3	TI SN74LS157N	Quad 2:1 multiplexer; channel 1 selects MANUAL or ASTABLE under control of U2 pin 6 (Q).

5. KiCad schematic baseline

The repository contains the Athena KiCad project at hardware/kicad/Athena/Athena.kicad_pro, with Clock.kicad_sch as the hierarchical clock sheet and hardware/kicad/Athena/exports/Athena-clock.pdf as the exported A3 schematic. The project record identifies KiCad 10.0.3 on macOS.

ATHENA CLOCK MODULE — KiCad schematic baseline (reproduced)

Repository topology: NE556 timing, SN7400 RUN/MANUAL and HLT logic, SN74LS157 selector

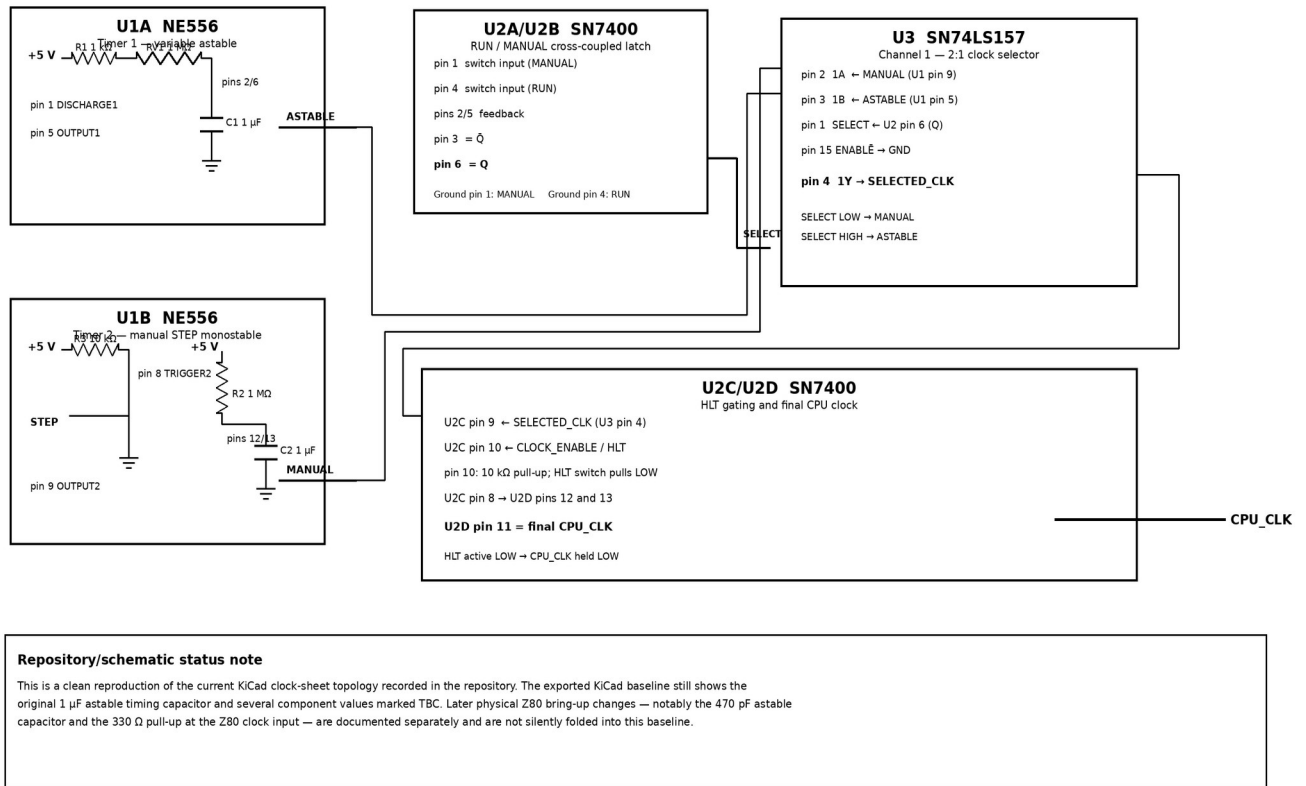


Figure 1. Current KiCad clock-sheet topology, reproduced from the repository schematic/export record.

Repository status: ERC was recorded as 0 errors and 1 warning; the warning is expected because CPU_CLK terminates at the module boundary while the receiving CPU lies outside the clock sheet.

Important limitation: The KiCad baseline still shows the original 1 µF astable timing capacitor and several component values marked TBC. Later physical Z80 bring-up changes are therefore documented separately rather than silently presented as if already incorporated into the schematic.

5.1 KiCad items still marked TBC or provisional

- RUN/MANUAL pull-up resistor values R4 and R5 were not yet confirmed in the schematic record.
- HLT and astable LED resistor values R7 and R8 remain to be confirmed.
- D2 is associated with the astable signal, but its complete LED wiring/polarity record was still provisional.
- The unused end of RV1 is drawn open; the physical breadboard should be checked to confirm whether it is strapped to the wiper.
- The physical terminal numbering/type of the RUN/MANUAL switch SW2 remains to be documented exactly.
- Local bypass capacitors C3, C4 and C5 are provisions in the drawing; fitted presence/values require confirmation.
- Unused SN74LS157 data inputs are intended to be held at a defined logic level; the current breadboard should be checked before the schematic is promoted to definitive as-built status.

6. U1A: variable-speed astable generator

Timer 1 of the NE555 is wired as an astable oscillator. The timing capacitor repeatedly charges and discharges between the internal approximately $1/3$ VCC and $2/3$ VCC comparator thresholds. The output at pin 5 changes state as the internal latch responds to those threshold crossings.

6.1 Baseline timing network

The repository baseline uses +5 V $\rightarrow$ 1 k Ω fixed resistor $\rightarrow$ pin 1 (DISCHARGE1) $\rightarrow$ 1 M Ω variable resistor $\rightarrow$ joined pins 2/6 $\rightarrow$ timing capacitor $\rightarrow$ GND. The original KiCad baseline shows a 1 μ F timing capacitor.

6.2 Oscilloscope verification

Pin	Function	Observed	Interpretation
1	DISCHARGE1	Vmin $\approx$ -80 mV; Vmax $\approx$ 5.12 V; Vpp $\approx$ 5.20 V; switching waveform	Discharge path is switching as the astable cycle changes state.
2	THRESHOLD1	Vmin $\approx$ 1.750 V; Vmax $\approx$ 3.370 V; Vpp $\approx$ 1.620 V; ramp	Timing node traverses the expected $\sim 1/3$ to $\sim 2/3$ VCC region.
3	CONTROL1	$\approx$ 3.29-3.37 V; $\sim$ 80 mV variation	Internal control reference sits near $2/3$ VCC.
4	RESET1	Held HIGH	Timer 1 reset is inactive; oscillator permitted to run.
5	OUTPUT1	Vmin $\approx$ -80 mV; Vmax $\approx$ 3.64 V; Vpp $\approx$ 3.72 V; square wave	Valid TTL-like astable source presented to the selector.
6	TRIGGER1	Vmin $\approx$ 1.76 V; Vmax $\approx$ 3.32 V; Vpp $\approx$ 1.56 V; ramp	Same timing node behaviour as pin 2, consistent with joined trigger/threshold node.
7	GND	Near ground; no switching function expected	Common 0 V reference.

7. U1B: manual STEP monostable

Timer 2 is a monostable one-shot. Its trigger input is normally held HIGH through 10 k Ω . Pressing STEP pulls pin 8 LOW. The internal latch then changes state, the discharge transistor releases the timing node, the capacitor charges through the 1 M Ω timing resistor, and the output pulse ends when the threshold comparator detects the timing node reaching its threshold.

7.1 Verified trigger and output behaviour

Pin	Function	Observed	Interpretation
8	TRIGGER2	Idle $\approx$ 4.72-5.04 V; pressed transition to $\sim$ 0 V; Vpp $\approx$ 5.44 V	STEP button provides the required active-LOW trigger.
9	OUTPUT2	Steady LOW $\approx$ 0 V; HIGH $\approx$ 4.32 V; edge overshoot observed	Manual one-shot output is present and suitable as the selector's A input.
12	THRESHOLD2 timing node	After correction: exponential rise with peak around 3.20 V, then rapid return near 0 V	Correct monostable RC timing behaviour restored.

13	DISCHARGE2 timing node	Externally the same RC node as pin 12	Internal discharge transistor controls the rapid return to ground.
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7.2 Major fault: wrong Timer 2 timing resistor

Expected: A 1 M Ω resistor from the Timer 2 timing node to +5 V, with a 1 μ F capacitor, giving a human-visible monostable interval.

Observed: The fitted resistor measured approximately 0.9943 k Ω , about 994 Ω .

Causal chain: Using $\sim$ 1 k Ω instead of 1 M Ω shortened the RC time constant by roughly 1000 $\times$. With 1 μ F, the pulse was around the millisecond scale, so on the previous slow oscilloscope timebase the capacitor charge looked almost instantaneous and the node appeared misleadingly square-like.

Correction: The $\sim$ 1 k Ω part was replaced with 1 M Ω .

Result: The expected exponential pin-12 charging ramp appeared immediately, peaking around 3.20 V before rapid discharge.

Learning: When a timing waveform contradicts the model, verify the physical R and C values and node connectivity before inventing a more complicated explanation.

8. U2A/U2B: RUN/MANUAL state latch

SN7400 gates 1 and 2 are cross-coupled to form a bistable NAND latch. Unlike the two NE556 sections, this part does not generate time. It stores the selected operating mode.

SN7400 pin	Role
1	Gate 1 switch input; normally HIGH, grounded to select MANUAL according to the schematic convention.
2	Feedback from pin 6 (Q).
3	Gate 1 output, $\bar{Q}$; feeds pin 5.
4	Gate 2 switch input; normally HIGH, grounded to select RUN according to the schematic convention.
5	Feedback from pin 3 ($\bar{Q}$).
6	Gate 2 output, Q; drives SN74LS157 SELECT pin 1.

The latch gives the switch action memory: after the appropriate input is asserted and released, the cross-coupled feedback keeps the selected state. This separates the act of choosing RUN or MANUAL from the clock waveforms themselves.

9. U3: SN74LS157 clock selector

Only channel 1 of the SN74LS157 is required for the clock. The device is permanently enabled by holding its active-LOW strobe at ground. Its common SELECT input is driven by the Q output of the SN7400 latch.

SN74LS157 pin	Connection	Meaning
1 SELECT	SN7400 pin 6 (Q)	LOW selects MANUAL; HIGH selects ASTABLE/RUN.
2 1A	NE556 pin 9	Manual STEP pulse.

3 1B	NE556 pin 5	Astable clock.
4 1Y	SN7400 pin 9	Selected clock passed to HLT gating.
8	GND	0 V reference.
15 ENABLE	GND	Selector permanently enabled.
16	+5 V	Supply.

$$Y = A \cdot \text{SELECT} + B \cdot \text{SELECT}$$

In words: the output follows A when SELECT is LOW, and follows B when SELECT is HIGH. Functionally, each selector channel performs the equivalent of two gated paths combined into one output. The actual LS-TTL silicon implements this Boolean function with transistor networks rather than literal miniature packaged gates.

10. U2C/U2D: HLT gating and final CPU clock

The selected clock from SN74LS157 pin 4 enters SN7400 gate 3 at pin 9. Pin 10 is CLOCK_ENABLE/HLT and is normally held HIGH through a 10 kΩ pull-up. The HLT switch pulls pin 10 LOW. Gate 3 then feeds both inputs of gate 4, which restores the intended output polarity.

SN7400 pin	Function
9	Selected clock input from SN74LS157 pin 4.
10	CLOCK_ENABLE/HLT; 10 kΩ pull-up, HLT switch to ground.
8	Gate 3 output; feeds both pin 12 and pin 13.
12, 13	Gate 4 inputs tied together to pin 8.
11	Final internal CPU_CLK output.

Verified behaviour: Pin 10 HIGH allows the selected clock to appear at pin 11 with the intended polarity; pin 10 LOW holds pin 11 LOW.

Indicator: The HLT LED is wired to indicate the LOW/active HLT state: +5 V → resistor → LED → pin 10.

11. Major end-to-end fault: selector bypassed

After the individual functional blocks existed, MANUAL mode still failed at the final output: the astable clock continued to appear even when manual operation was selected. The eventual fault was architectural, not inside the 556, latch or multiplexer.

Observed: SN7400 pin 9 was still directly connected to NE556 pin 5, the astable output.

Expected architecture: SN7400 pin 9 should receive only the selected clock from SN74LS157 pin 4.

Causal chain: The legacy direct connection bypassed the selector. Regardless of the RUN/MANUAL latch and the SN74LS157 output, the astable clock still drove the HLT-gating input directly.

Correction: The direct NE556 pin 5 → SN7400 pin 9 connection was removed and SN74LS157 pin 4 → SN7400 pin 9 was installed.

Observed result: RUN, MANUAL/STEP and HLT then behaved correctly at the final SN7400 pin-11 CPU clock output.

Historical hypothesis: The direct 556-to-7400 connection was probably left over from the earlier astable-plus-HLT stage before the selector was inserted. This is a plausible reconstruction, not a documented fact.

12. End-to-end functional verification

Mode / control	Expected CPU_CLK behaviour	Verified result
RUN selected; HLT inactive	Continuous variable-speed astable clock	Observed at final output.
MANUAL selected; no STEP	No continuous clock	Observed after selector-bypass fault was corrected.
MANUAL selected; STEP pressed	Single monostable pulse appears at CPU_CLK	Observed.
HLT active	CPU_CLK held LOW regardless of selected source	Observed.
HLT released	Selected source again reaches CPU_CLK	Observed.

At this point the clock module was functionally complete for its intended architecture. The result was not merely that pin 11 toggled: the individual sources, stored mode, selector and halt gate had each been understood and exercised.

13. Later Z80 bring-up adaptations

The first CPU load exposed two additional requirements: a faster practical operating range and a higher clock-HIGH level at the NMOS Z80 input. These were treated as adaptations at the operating-range/interface boundary rather than reasons to replace the clock architecture.

13.1 Astable timing capacitor change

Baseline: The original clock schematic uses a 1 μ F astable timing capacitor for very slow, easily observed operation.

Experiment: A 1 nF capacitor (code 102) was tried; one measured operating point was approximately 12.37 kHz with $V_{min} \approx 240$ mV, $V_{max} \approx 5.12$ V and $V_{pp} \approx 4.88$ V.

Current Z80 bring-up value: The astable timing capacitor was changed to 470 pF (code 471).

Observed: The Z80 clock was subsequently measured around 286.7-288.5 kHz at the selected potentiometer setting.

13.2 NMOS Z80 clock-level adaptation

Observed before adaptation: SN7400 pin 11 / Z80 pin 6 reached only about 3.8 V HIGH under the original TTL drive.

Documented interface issue: The older NMOS Z80 clock input has a special clock-HIGH requirement close to VCC, so a normal TTL HIGH does not provide the desired guaranteed margin.

Change: A 330 Ω pull-up was added from +5 V to the Z80 pin-6 clock node. The SN7400 pin-11 output remains directly connected to the same node; the resistor is not in series.

Observed after adaptation: At the Z80 clock pin the waveform reached approximately 5.04-5.12 V HIGH with a LOW around 0.08-0.24 V in later captures.

Boundary: The 330 Ω pull-up belongs to the Z80 interface arrangement, not to the original clock-module KiCad schematic.

13.3 Manual single-T-state stepping capability

The already-verified MANUAL path has an important consequence once the clock is connected to the Z80: each STEP press produces one clock pulse, so the processor advances by one T-state rather than by one complete instruction.

This makes it possible to walk deliberately through an instruction machine cycle and observe when the address bus, data bus and control signals change. Signals of particular interest include A0-A15, D0-D7, M1, MREQ, RD, WR, RFSH, IORQ and CLK.

Established electrical behaviour: MANUAL mode selects the NE556 monostable and one STEP press produces one pulse at the final CPU_CLK output.

CPU-debugging consequence: With the Z80 connected, one pulse advances one T-state. This is a consequence of the clocked CPU architecture and gives the existing clock module a single-cycle observability function.

Use: Compare each manually advanced T-state with the Z80 timing diagrams and the simultaneously observed bus/control signals. The feature is cycle-level stepping and must not be confused with instruction-level single-step debugging.

14. Technical learning

14.1 Timing is an internal analogue process producing a digital result

The NE556 measurements made the internal mechanism visible: the timing capacitor moves continuously between threshold voltages while comparator, latch and discharge-transistor actions convert that analogue ramp into a digital output waveform.

14.2 A latch stores state rather than generating time

The SN7400 RUN/MANUAL pair introduced a different kind of logic. The two NAND gates use feedback to preserve a binary state after the switch action has ended. This state then controls another device rather than becoming the clock directly.

14.3 A multiplexer is controlled routing

The SN74LS157 does not create either clock source. It routes one of two already-existing signals according to the stored SELECT state. This distinction between generation and selection was essential to understanding the selector-bypass fault.

14.4 Logical correctness does not guarantee electrical compatibility

The clock worked correctly as TTL logic before it was attached to the NMOS Z80, yet the CPU clock input imposed a stricter HIGH-level requirement. The later pull-up is therefore an interface correction, not evidence that the clock's Boolean design was wrong.

14.5 Observability changes debugging quality

Because ASTABLE, MANUAL, Q, SELECTED_CLK, CLOCK_ENABLE and CPU_CLK are individually observable, faults can be localised stage by stage. The clock became a chain of testable hypotheses rather than a single 'works/does not work' object.

15. Fault-isolation lessons

- Measure the component actually fitted, not the value expected from the schematic.
- Verify switch and breadboard connectivity from physical evidence rather than assumed topology.
- When a downstream signal is wrong, first verify the upstream signal at each architectural boundary.
- A circuit can contain individually correct blocks and still fail because one legacy wire bypasses the intended signal path.
- Oscilloscope timebase matters: a millisecond-scale monostable viewed at a much slower scale can create a misleading visual impression.
- Small edge overshoot is an observation to characterise, not automatically a fault to assign. Probe-ground inductance, breadboard parasitics and real output behaviour remain candidate causes until discriminated experimentally.
- Do not redesign a verified subsystem merely because a later load introduces a new interface requirement; adapt the boundary when that is the actual problem.

16. Current status and confidence

Area	Status	Evidence / qualification
NE555 astable generation	Confirmed	Threshold/trigger ramps and output waveform measured.
NE555 manual monostable	Confirmed	Trigger, output and timing ramp observed after resistor correction.
RUN/MANUAL latch	Confirmed functional	Persistent selection controls SN74LS157 SELECT as intended.
SN74LS157 selection	Confirmed functional	Manual versus astable selection verified after bypass wire removed.
HLT gating	Confirmed	HLT LOW holds final clock LOW; release restores selected source.
Final CPU_CLK	Confirmed	SN7400 pin 11 provides final internal clock.
KiCad core connectivity	Confirmed baseline	Repository netlist/export checked; ERC 0 errors, one expected module-boundary warning.
KiCad as-built completeness	Provisional	Several resistor/LED/bypass/switch details remain TBC; later 470 pF change not yet folded into baseline.
Z80 frequency adaptation	Confirmed physical	470 pF timing capacitor gives current few-hundred-kHz operating range.
Z80 clock-level adaptation	Confirmed physical	330 Ω pull-up at CPU clock node raises HIGH level close to +5 V.

17. Next documentation and design updates

1. Use MANUAL/STEP during Z80 timing work to capture and document individual T-states against the processor timing diagrams, preserving this as a deliberate observability feature of the clock architecture.
2. Update the KiCad astable timing capacitor to reflect the physical 470 pF value if that is retained as the current configuration, or represent selectable clock ranges explicitly if that becomes the design direction.
3. Document the Z80 clock-interface pull-up separately at the CPU-board/interface boundary rather than burying it inside the generic clock module.
4. Resolve the KiCad TBC values for RUN/MANUAL pull-ups, LED resistors, local bypass capacitors and switch terminal numbering from fresh physical evidence.
5. Verify the physical treatment of unused SN74LS157 inputs and the RV1 unused terminal before declaring the schematic definitive as-built.
6. Archive the superseded individual-555 implementation as foundation-course evidence rather than deleting it.
7. Continue using the clock as a known reference subsystem while the Z80 bus, memory and firmware stages are developed.

Appendix A. NE556 pin map used by the clock

Pin	Function in Athena
1	DISCHARGE1: astable timing network.
2	THRESHOLD1: joined to pin 6 and astable timing capacitor node.
3	CONTROL1: no functional external control connection in baseline.
4	RESET1: +5 V.
5	OUTPUT1: ASTABLE → SN74LS157 pin 3.
6	TRIGGER1: joined to pin 2 timing node.
7	GND.
8	TRIGGER2: 10 k Ω pull-up; STEP switch pulls LOW.
9	OUTPUT2: MANUAL → SN74LS157 pin 2.
10	RESET2: +5 V.
11	CONTROL2: no functional external control connection in baseline.
12	THRESHOLD2: manual monostable timing node.
13	DISCHARGE2: same external timing node; internal discharge transistor.
14	VCC: +5 V.

Appendix B. SN7400 pin map used by the clock

Pin	Function in Athena
1	Latch gate 1 input; switch-driven mode input.
2	Latch feedback from pin 6.
3	Latch complementary output $\bar{Q}$ → pin 5.
4	Latch gate 2 input; switch-driven mode input.
5	Latch feedback from pin 3.
6	Latch Q → SN74LS157 SELECT pin 1.
7	GND.
8	HLT gate 3 output → pins 12 and 13.
9	Selected clock input ← SN74LS157 pin 4.
10	CLOCK_ENABLE/HLT; 10 k Ω pull-up, switch to GND.
11	Final CPU_CLK.
12	Gate 4 input tied to pin 8.
13	Gate 4 input tied to pin 8.
14	VCC +5 V.

Appendix C. SN74LS157 channel-1 map

Pin	Function in Athena
1	SELECT ← SN7400 pin 6 (Q).
2	1A ← NE556 pin 9 MANUAL.
3	1B ← NE556 pin 5 ASTABLE.
4	1Y → SN7400 pin 9 SELECTED_CLK.
8	GND.
15	ENABLE → GND.
16	+5 V.

Appendix D. Evidence classification

Classification	Meaning
Observed fact	Directly measured or visibly demonstrated on the physical breadboard/instrument.
Documented fact	Recorded in the repository, schematic or component documentation.
Deduction	A conclusion supported by observations plus the working circuit model.
Hypothesis	A plausible explanation that still requires a discriminating test.
Provisional / TBC	Intentionally unresolved and not to be promoted silently into a permanent design decision.

Appendix E. Repository records consulted

- hardware/breadboard/clock/wiring.md
- docs/design-decisions/clock-architecture.md
- docs/measurements/clock-556-verification.md
- docs/faults/clock-fault-history.md
- docs/milestones/clock-module.md
- hardware/breadboard/clock/README.md
- hardware/kicad/Athena/README.md
- hardware/kicad/Athena/exports/Athena-clock.pdf